

Math 526, S12
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	22	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ where R denotes the reduced echelon form of A .

$$[A \ b] = \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 8 & 10 & 2 \\ 3 & 6 & 7 & 10 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & -2 & -2 & -2 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 0 & 1 & -3 \\ 0 & 0 & \textcircled{1} & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] = [R \ d]$$

(b) Find a particular solution x_p of $Ax = b$ by setting the free variables to be zero.

$$Ax_p = b$$

free variables: x_2, x_4 .

$$\begin{cases} x_1 + 2x_2 + x_4 = -3 \\ x_3 + x_4 = 1 \end{cases}, \text{ By setting } x_2 = x_4 = 0 \Rightarrow \begin{cases} x_1 = -3 \\ x_3 = 1 \end{cases}$$

$$\Rightarrow x_p = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

(c) Find the complete solution to $Ax = b$.

Special solutions: $AX_h = 0 \Leftrightarrow RX = 0$

$$\begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

i) Set $x_2 = 1, x_4 = 0$. We get $x_1 = -2, x_3 = 0$

$$\Rightarrow s_1 = (-2, 1, 0, 0)$$

ii) Set $x_2 = 0, x_4 = 1$. We get $x_1 = -1, x_3 = -1$

$$\Rightarrow s_2 = (-1, 0, -1, 1)$$

Thus,

$$x = x_p + c_1 s_1 + c_2 s_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Problem 2. (20 points)

Let $A = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2 \quad \leftarrow \text{Pivot variables: } x_1, x_3.$$

$$\text{a basis: } \left\{ \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2$$

$$\text{a basis: } \left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the nullspace of A , $N(A)$.

free variables: $x_2, x_4 \Rightarrow \dim(N(A)) = 2$
 We solve $Ax = 0$, i.e., $\begin{cases} 2x_1 + 4x_2 + 5x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$

i) Set $x_2 = 1, x_4 = 0$, we get $x_1 = -2, x_3 = 0$

$$\Rightarrow s_1 = (-2, 1, 0, 0)$$

ii) Set $x_2 = 0, x_4 = 1$, we get $x_1 = -\frac{5}{2}, x_3 = -2$

$$\Rightarrow s_2 = (-\frac{5}{2}, 0, -2, 1)$$

$$\Rightarrow \text{a basis of } N(A) : \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$N(A^T) = 3 - \dim(C(A^T)) = 3 - 2 = 1$$

$$\text{Solve } A^T y = 0 \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 2 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{cases} 2y_1 = 0 \\ 4y_1 = 0 \\ y_2 = 0 \\ 5y_1 + 2y_2 = 0 \end{cases} \Rightarrow y_1 = y_2 = 0. \Rightarrow y = \begin{bmatrix} 0 \\ 0 \\ y_3 \end{bmatrix} = y_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{a basis of } N(A^T) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Problem 3. (20 points)

(a) Find the straight line $C + Dt$ that comes closest to the values $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting.

Solution: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$, $x = \begin{bmatrix} C \\ D \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \quad A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

Solve $A^T A x = A^T b$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\Rightarrow C = 1, D = 4$$

$\Rightarrow$ The least squares line is $y = 1 + 4t$.

(b) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Find A^{-1} using the cofactor formula $C^T / \det A$.

Solution: $\det A = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = (-1) \cdot (-1) + 3 \cdot 1 = 4$

$$C = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -6 & 2 \\ 1 & 3 & -1 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow A^{-1} &= \frac{C^T}{\det A} = \frac{1}{4} \begin{bmatrix} -1 & 2 & 1 \\ 1 & -6 & 3 \\ 1 & 3 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -1/4 & 1/2 & 1/4 \\ 1/4 & -3/2 & 3/4 \\ 1/4 & 1/2 & -1/4 \end{bmatrix} \end{aligned}$$

Problem 4. (22 points)

(a) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Find an orthonormal basis q_1, q_2, q_3 for the column space of A using the **Gram-Schmidt** process.

Solution: $\tilde{q}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 6$

$$\tilde{q}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{\tilde{q}_1 \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{\|\tilde{q}_1\|^2} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{0}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_2\| = 2$$

$$\begin{aligned} \tilde{q}_3 &= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{\tilde{q}_1 \cdot \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}}{\|\tilde{q}_1\|^2} \tilde{q}_1 - \frac{\tilde{q}_2 \cdot \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}}{\|\tilde{q}_2\|^2} \tilde{q}_2 \\ &= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{9}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \|\tilde{q}_3\|^2 = 3 \end{aligned}$$

$$\Rightarrow q_1 = \tilde{q}_1 / \|\tilde{q}_1\| = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}, \quad q_2 = \tilde{q}_2 / \|\tilde{q}_2\| = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$q_3 = \tilde{q}_3 / \|\tilde{q}_3\| = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

(b) Find the area of the triangle formed by the three points $(-1, 2)$, $(0, 5)$, $(3, 3)$ using the determinant formula.

Solution: $\det \begin{pmatrix} -1 & 2 & 1 \\ 0 & 5 & 1 \\ 3 & 3 & 1 \end{pmatrix} = -1 \cdot (5 \cdot 1 - 3 \cdot 1) + 3(2 \cdot 1 - 5 \cdot 1)$

$$\begin{aligned} &= -2 - 9 \\ &= -11 \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Area} &= \frac{1}{2} |\det(A)| \\ &= \frac{1}{2} \cdot |-11| \\ &= \frac{11}{2} \end{aligned}$$

Problem 5. (18 points) True or False (no explanation needed).

(a) If the columns of a matrix are independent, so are the rows.

False

(b) If $\mathbf{A}^T = -\mathbf{A}$ then the row space of $\mathbf{A}$ equals the column space of $\mathbf{A}$.

True

(c) The nullspace $N(\mathbf{A})$ and the row space $C(\mathbf{A}^T)$ are orthogonal complements.

True

(d) If $\mathbf{P}$ is a projection matrix, then $\mathbf{P}^2 = \mathbf{P}$.

True

(e)
$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & e & f \\ -a & b & c \\ -g & h & i \end{vmatrix}.$$

True

(f) For the vectors of same dimension $\mathbf{u}, \mathbf{v}, \mathbf{w}$, we have that $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{u}) \cdot \mathbf{w}$.

False